

AI Availability as a Third Measurable Layer of Brand Availability

Five-Substrate Empirical Anchoring of the AIAS™ Presence Measurement Protocol — Synthesis
under Locked v1.6 Methodology

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Abstract

The AIAS™ Presence Measurement Protocol operationalizes AI Availability — the brand-level probability that an AI intermediary retrieves, recommends, or selects a brand in a category-anchored decision context — as a third measurable layer of brand availability alongside Ehrenberg-Bass Mental Availability and Physical Availability. The construct is claimed as measurable under pre-registration discipline against a five-substrate empirical anchor base.

The anchor base spans kitchenware (v0.16 + v0.17), indie fragrance (v0.18), audiophile headphones (v0.19), skincare (v0.20), and cosmetics (v0.21), measured against a six-LLM reference panel held fixed across phases. Protocol v1.6 consolidates the methodology with three increments — substrate Recognition pre-screen, independent Identity-Load moderator pathway, and Phantom Brand Persistence Phase B extension — and locks the retrospective-scoring boundary at v1.6's specification: v0.20 and v0.21 score under v1.6 where each increment is supported; v0.16–v0.19 stand on their published v1.4-era verdicts.

Headline findings at v0.21: Type 2 EMERGED (three Cell B cases, with Rare Beauty's $R_{cat} = 1 / R_{cult} = 17$ anchoring the textbook Recognition $\times$ Recall dissociation); $H_{IdentityLoad_Direct}$ CONFIRMED (Cell B $\delta = +7.13$ with 95% bootstrap CI [+4.75, +10.25], the program's first CONFIRMED moderator verdict at any layer); $H_{PhantomBrandPersistence}$ CONFIRMED (six off-panel brands clear $K = 6$, validity anchor passes with margin). A cross-phase synthesis-layer contribution — out-of-cell Type 2 in La Mer (v0.20 Cell A) and e.l.f. Cosmetics (v0.21 Cell C) — establishes the dissociation pattern's recurrence beyond per-phase Cell-B-anchored verdict matrices.

AIAS™ 1.0 names the Presence component of the multi-component AIAS construct; the full six-component composite is a multi-year roadmap and is not claimed here. Construct validity and the consumer-behavior correlate of AI Availability are held as Phase 3 future work. All pre-registrations, acquisition data, scoring code, and verdict matrices are deposited at osf.io/ec6wh.

1. Introduction

The mediation of category-anchored brand retrieval by large language model (LLM) intermediaries is now a structural feature of consumer discovery. A consumer who asks an AI assistant for the best knife, a popular fragrance, a viral cosmetics brand, or a quality skincare routine receives an answer drawn from the intermediary's training distribution, retrieval substrate, and response policy — not from the consumer's own associative memory and not from the brand's physical distribution footprint. The discovery surface has acquired a third layer alongside the two layers the Ehrenberg-Bass tradition has formalized: Mental Availability, the probability that a brand is retrieved from category-anchored memory at decision time (Sharp,

2010; Romaniuk, 2018); and Physical Availability, the probability that a brand is accessible at decision time through retail, digital, or distribution channels (Sharp & Romaniuk, 2021). This paper presents the foundational claim that AI Availability — the brand-level probability that an AI intermediary retrieves, recommends, or selects a brand in a category-anchored decision context — is a third measurable layer of brand availability, and presents the AIAS™ Presence Measurement Protocol as the operational instrument for measuring it.

The claim is bounded. AIAS™ 1.0 names the Presence component of a multi-component AIAS construct; the full composite is a multi-year research program (González Castro, 2026a), and only Presence is currently operationalized. The claim is also a measurability claim, not a construct-validity claim: the synthesis establishes that AI Availability can be measured under a falsifiable pre-registered protocol with cross-substrate replication, leaving predictive validity against behavioral outcomes, convergent validity across panel constructions, and discriminant validity against Mental and Physical Availability as future work. The synthesis is silent on the consumer-behavior correlate of AI Availability; that correlation is not yet established and is held explicitly out of scope.

The empirical anchor base for the claim spans five substrate families acquired between v0.16 (kitchen knives; González Castro, 2026g) and v0.21 (cosmetics; González Castro, 2026l): kitchenware (v0.16 and v0.17, indexed across knife and broader premium tiers), indie fragrance (v0.18), audiophile headphones (v0.19), skincare (v0.20), and cosmetics (v0.21). Each phase is a pre-registered measurement event against a panel of six LLMs held fixed across the program. The cumulative six-phase corpus is consolidated under a single locked methodology version — Protocol v1.6 (González Castro, 2026f) — which introduces a substrate-level Recognition pre-screen, an Identity-Load moderator pathway evaluable independently of the protocol's Regime 4 conditions, and a Phantom Brand Persistence Phase B extension that converts a previously discovery-level observation into a measured component. The synthesis consumes v1.6's retrospective scoring boundaries without extending them: where the data permits scoring under v1.6 (v0.20 and v0.21), v1.6 is used; where it does not (v0.16–v0.19), the published verdicts from each phase paper stand and are joined to the synthesis by narrative classification for Increment 1 only.

The construct claim is presented in seven layers. AI Availability sits as a third measurable layer alongside Mental and Physical Availability (Claim L1, §5.1). The construct decomposes into at least two empirically separable components, Recognition and Recall (L2, §5.2; González Castro, 2026d). Recall decomposes further into a canonical-authority channel (R_cat) and a cultural-footprint channel (R_cult) (L3, §5.3; González Castro, 2026e). Three canonical patterns occupy the Recognition × Recall plane — Iwachu, Type 1, and Type 2 (L4, §5.4). Per-cell channel asymmetry tracks the Identity-Load gradient of the substrate's cell construction (L5, §5.5; González Castro, 2026f). Off-panel brand mentions persist under panel-anchored retrieval as an empirical regularity (L6, §5.6). And the present paper claims measurement of the Presence com-

ponent only, naming the full multi-component composite as a multi-year roadmap rather than a present specification (L7, §5.7).

The remainder of the paper is structured as follows. §2 positions AI Availability theoretically against the Ehrenberg-Bass canon and the Tri-System framework (González Castro, 2026a). §3 consolidates the v1.2 → v1.6 methodology chain that anchors the measurement procedure. §4 presents the empirical evidence across the five substrate families, with one subsection per family. §5 develops the seven-layer construct claim. §6 discusses implications for marketing science and brand practice and identifies the limitations of the present synthesis. §7 outlines the future-research program — construct validation, the full six-component composite, substrate expansion, and cross-language replication. The paper closes with references and declarations.

2. Theoretical positioning

The Ehrenberg-Bass tradition formalizes brand availability as two structurally distinct probabilities. Mental Availability is the probability that a brand is retrieved from memory in a category-anchored decision context. Operationalized through category entry points, distinctive brand assets, and memory-network strength, it has been the central empirical construct of the discipline for two decades (Sharp, 2010; Romaniuk, 2018). Physical Availability is the probability that a brand is encounterable at decision time through the distribution footprint that connects the brand to the consumer’s purchase environment, including retail shelves, digital storefronts, and channel partners. The two are treated as orthogonal: a brand can be mentally available without being physically available in a given market, or physically available without sufficient memory anchoring to be considered (Sharp & Romaniuk, 2021).

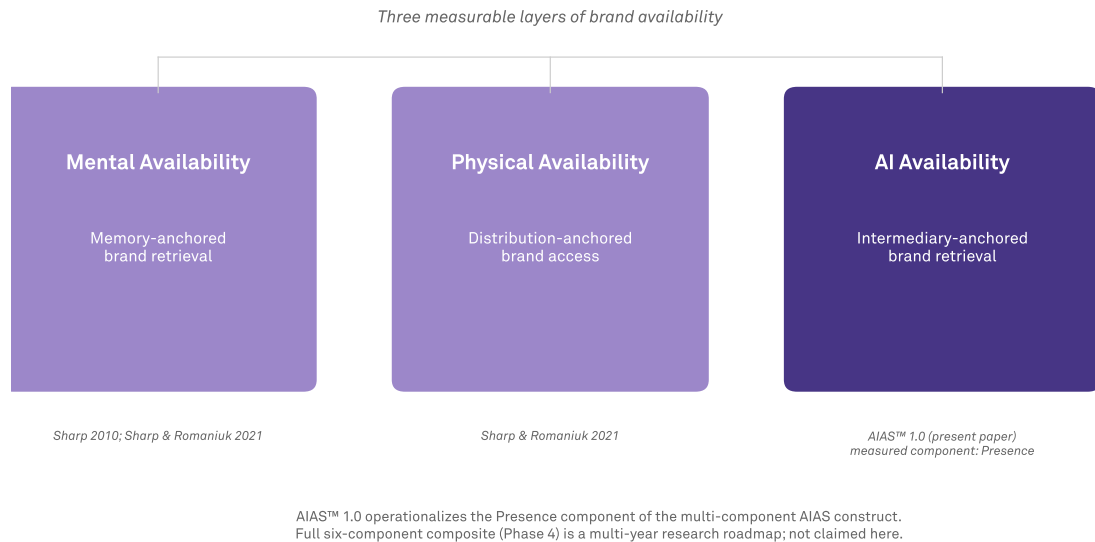
AI-mediated retrieval breaks the two-layer framing in a structural sense. When the consumer’s decision surface is an LLM-mediated conversation, the brand is neither retrieved from the consumer’s own memory nor encountered through the brand’s distribution footprint. It is retrieved by an intermediary whose own representational state is a function of training data, retrieval policy, and the prompt context — none of which is reducible to consumer memory or to physical-distribution presence. The intermediary’s selection of a brand for a category-anchored response is a third event in the availability chain, located between Mental Availability (which it does not require) and Physical Availability (which it does not require either). The construct that captures this third event is AI Availability.

The Tri-System Brand Growth framework (González Castro, 2026a) positions AI Availability as the third system of brand presence alongside Mental and Physical Availability and provides the architectural context for the present synthesis. Tri-System frames the full multi-component AIAS composite (a 0–100 score across six measurable dimensions — Presence, Ranking, Consistency, Coverage, Grounding, Sentiment) as a multi-

year research arc. AIAS™ 1.0 — the present paper — operationalizes the first of those six dimensions: Presence, the brand-level probability that an AI intermediary retrieves, recommends, or selects a brand at all in a category-anchored decision context. The remaining five dimensions ship under a Phase 4 successor program; their specification is reserved and not load-bearing for the present construct claim.

Three measurable layers of brand availability

AI Availability as a third measurable layer alongside Ehrenberg-Bass Mental and Physical Availability



Source: AIAS™ 1.0 synthesis paper data — osf/aiaas_1.0/aiaas_1.0_synthesis_data.json (lock tag aiaas-1-0-data-locked). Upstream: SSRN 6791999 (v0.16), 6802261 (v0.17), 6806558 (v0.18), 6809182 (v0.19), 6811441 (v0.20), 6815378 (v0.21); methodology v1.6 SSRN 6816340. Five methodology papers v1.2–v1.6 in citation chain.

Figure 1: The three measurable layers of brand availability: Mental Availability (Sharp, 2010; Romaniuk, 2018), Physical Availability (Sharp & Romaniuk, 2021), and AI Availability (the present paper, operationalizing Tri-System’s third-system layer at the Presence component).

Two demarcations are essential. First, the synthesis claims measurability under a falsifiable protocol, not construct validity. Construct validity — predictive validity against behavioral outcomes (purchase, consideration, search), convergent validity across alternative panel constructions, and discriminant validity against Mental Availability and Physical Availability — is the program’s Phase 3 work. The present paper establishes that AI Availability behaves as a measurable property of brand-substrate-intermediary triples under pre-registration discipline; whether the measurement predicts downstream consumer behavior is a separate empirical question. Second, the synthesis is silent on the consumer-behavior correlate of AI Availability. The correlation is plausible on theoretical grounds and is implied by the Ehrenberg-Bass logic that availability constructs are load-bearing for category buying — but the present program has not measured it, and

over-claim on this point would compromise the program’s standing on the measurability claim that is established.

The positioning has a third implication for the paper’s argumentative scope. The synthesis is a measurement-program paper, not a comparative-LLM-performance paper. The six-slot reference panel held constant across phases is treated in aggregate; per-LLM rankings and provider-level performance comparisons are not the claim. The construct is a property of the brand $\times$ substrate $\times$ panel system as a whole, and reads accordingly: where v0.21 cosmetics produced Type 2 EMERGED, that verdict is a property of the cosmetics-IL-gradient substrate measured against the panel, not a claim about any particular LLM provider.

3. Methodology

The measurement procedure is the same across all six phases. The reference panel is a six-slot LLM panel locked at v0.17 and held constant across the program: Claude Opus 4.5, Claude Sonnet 4.5, GPT-4o, GPT-4o-mini, Gemini 2.5 Flash, and Gemini 2.5 Flash Lite (González Castro, 2026h). Holding the panel fixed across phases is the program’s cross-phase comparability discipline; verdicts are properties of brand $\times$ substrate $\times$ panel triples evaluated under the same set of intermediaries each time, and panel substitution is reserved for a methodology-version event rather than a phase-level edit.

Phase A is the Recognition layer. For each (brand $\times$ model) pair the model is prompted with a category-membership probe (template locked at the v1.4 specification); the per-brand Recognition score C_P is the count of “yes” responses across the six panel models, range 0–6 (González Castro, 2026d). C_P is the construct’s discrimination signal at the category-membership boundary: a brand with $C_P = 0$ is not recognized as a category member by any panel intermediary; a brand with $C_P = 6$ is fully canonized.

Phase B is the Recall layer. Each panel model receives a six-frame open-ended query battery against the substrate’s category, and the panel’s responses are scanned for brand mentions against the locked substrate registry. v1.5 introduced the two-channel decomposition: three canonical-authority frames (best, recommended, highest-quality) anchor R_cat (max 18: 3 frames $\times$ 6 models); three cultural-footprint frames (popular, celebrity, viral / cult) anchor R_cult (max 18) (González Castro, 2026e). Mention detection follows the v1.4 canonical-detection rules (case-insensitive, accent-stripped, possessive-aware, word-boundary anchored).

The dissociation taxonomy organizes the Recognition $\times$ Recall space into three named patterns. Iwachu — high Recognition with sparse canonical Recall ($C_P \geq 5 \wedge R_cat \leq 2$) — was the first dissociation case named in the program (González Castro, 2026d); Type 1 (canonical-preferred: $R_cat \geq 5 \wedge R_cult \leq 2$) and Type 2 (cultural-preferred: $R_cat \leq 2 \wedge R_cult \geq 5$) extend the framework into the v1.5 two-channel Recall plane (González Castro, 2026e). The three patterns are the construct-level vocabulary used in §4 substrate

narratives and in §5's claim L4.

Protocol v1.6 (González Castro, 2026f) closes the methodology layer for the present synthesis with three increments. Increment 1 is a substrate-level Recognition pre-screen: a substrate with distinct C_P count = 1 in all three cells is classified as uniform-saturation and bypasses the v1.5 C2 routing, collapsing the cell-level Regime 4 verdicts into a single substrate-level state, REGIME-4-UNAVAILABLE-AT-RECOGNITION, distinct from FALSIFIED. Increment 2 introduces H_IdentityLoad_Direct, a parallel hypothesis evaluable independently of Regime 4: per cell, $\delta = \text{mean}(R_{\text{cult}}) - \text{mean}(R_{\text{cat}})$ is computed with a 95% percentile bootstrap confidence interval ($n = 10,000$), and the verdict matrix is anchored to the IL-gradient prediction with a monotonic-gradient check on Cell C. Increment 3 lifts Phantom Brand Persistence from a descriptive observation to a measured component: R_phantom counts off-panel reference-vocabulary mentions across the 36 Phase B responses against a substrate-specific vocabulary constructed as the union of the panel, a pre-registered top-50 market-share list, and prior-phase Phase B emergents cross-validated by the market-share list, with persistence threshold $K = 6$.

The retrospective-scoring boundary for the present synthesis is locked at v1.6 §7 and is not extended here. Increment 1 applies retrospectively as narrative classification for v0.16–v0.19 (computable from each phase's published Phase A distribution) and as scored classification for v0.20 and v0.21. Increment 2 applies retrospectively to v0.20 and v0.21 only; v0.16–v0.19 lack R_cult — the two-channel decomposition was a v1.5 prospective increment first acquired at v0.20, and no defensible retrospective channel re-mapping is available. Increment 3 applies retrospectively to v0.21 only — the substrate where the construct is empirically motivated and where the v0.20 Phase B logs supply the cross-validated-emergent input. For v0.16–v0.19, each phase's published v1.4-era verdicts stand as the authoritative within-phase result; §4 surfaces them as the per-phase reference and adds the v1.6 narrative Increment 1 classification.

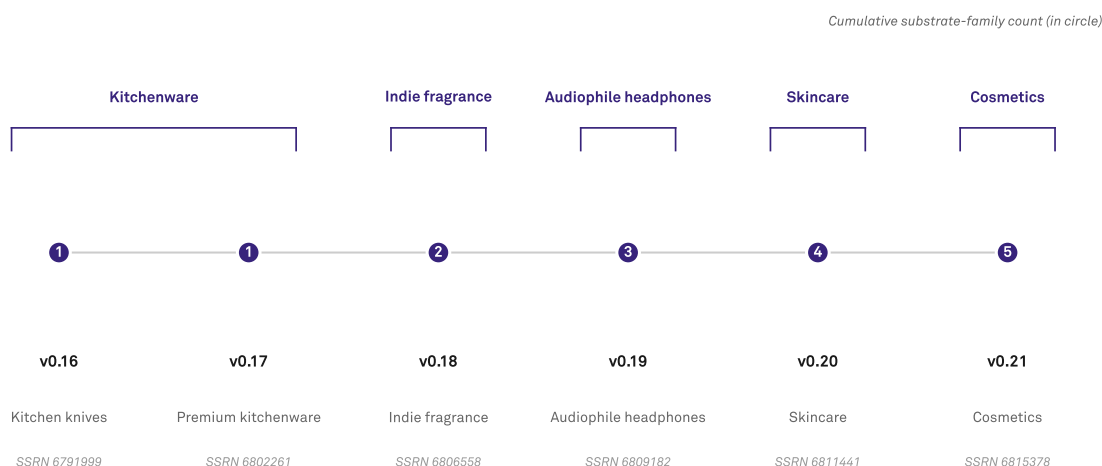
Every phase enters the program under pre-registration discipline (Nosek et al., 2018). Panel construction, substrate registry, Phase A probe template, Phase B six-frame battery, hypothesis set, decision rules, and verdict matrices are locked ex-ante at a git tag (v0.NN-prereg-rN); the methodology version is locked at its own tag (v1.NN-prereg-rN). All artifacts — pre-registrations, acquisition data, scoring code, verdict outputs, and reference vocabularies — are deposited under Open Science Framework project ec6wh (osf.io/ec6wh). The present synthesis is locked at aias-1-0-outline-locked (outline), aias-1-0-data-locked (cross-phase source-of-truth data), and aias-1-0-charts-locked (figures), with the final paper-lock tag cut at submission.

4. Empirical evidence across five substrate families

The five-substrate anchor base spans kitchenware (v0.16 and v0.17), indie fragrance (v0.18), audiophile headphones (v0.19), skincare (v0.20), and cosmetics (v0.21). Each family is presented at comparable depth in §4.1 through §4.5 below, with the same shape: substrate definition and cell construction; headline empirical findings under the methodology version prevailing at acquisition; v1.6 retrospective increment classification; and the construct contribution carried into §5. The kitchenware family establishes the Iwachu dissociation pattern, the original Regime 4 boundary, and the program's substrate-language-carryforward methodology object. Indie fragrance extends the dissociation framework to an English-language perfumistas-native substrate and articulates the IL-gradient moderator hypothesis as a coherent cross-phase question. Audiophile headphones is the substrate-shape case — a uniform-Identity-Load substrate that the IL-gradient framework cannot test against and that motivates the v1.6 Increment 1 substrate-level pre-screen. Skincare is the first prospective phase under v1.5's two-channel Recall design and produces the first PARTIAL Type 2 outcome under the v1.5 verdict matrix. Cosmetics is the program's strongest empirical anchor: Type 2 EMERGED, IL Direct CONFIRMED, Phantom Brand Persistence CONFIRMED, and the first measured uniform-saturation Recognition state. The retrospective-scoring boundary stated in §3 applies throughout: v0.20 and v0.21 are scored under v1.6 where each increment supports it (Increment 1 always; Increment 2 wherever R_cult is available; Increment 3 for v0.21 only), and v0.16–v0.19 stand on their published v1.4-era verdicts as the authoritative within-phase results, with v1.6 Increment 1 narrative classification added where each phase's published Phase A distribution supports it. Two cross-phase synthesis-layer contributions surface in §4.4 and §4.5 and are flagged at each occurrence: the all-cell algorithmic dissociation framework, applied uniformly to v0.20 and v0.21 data, recovers Type 2 cases in non-Cell-B positions that each phase's Cell-B-anchored verdict matrix correctly did not surface within its own scope. Figure 2 shows the cumulative substrate-family anchor base across the six phases.

Cumulative substrate-family anchor base (v0.16 → v0.21)

Five substrate families anchored under locked v1.6 methodology



Source: AIAS™ 1.0 synthesis paper data — osf/aias_1.0/aias_1.0_synthesis_data.json (lock tag aias-1-0-data-locked). Upstream: SSRN 6791999 (v0.16), 6802261 (v0.17), 6806558 (v0.18), 6809182 (v0.19), 6811441 (v0.20), 6815378 (v0.21); methodology v1.6 SSRN 6816340. Five methodology papers v1.2–v1.6 in citation chain.

Figure 2: Cumulative substrate-family anchor base across the six phases v0.16–v0.21. Five substrate families anchored under locked Protocol v1.6 methodology.

4.1 Kitchenware (v0.16 + v0.17)

The kitchenware family anchors the program in two pre-registered phases against substrates organized by heritage-tier (Cell A), Japanese-cult-tier (Cell B), and mass-market-tier (Cell C). v0.16 sampled kitchen knives (González Castro, 2026g); v0.17 extended the substrate to broader premium kitchenware (González Castro, 2026h). Kitchen knives are a useful program-opening substrate: the category carries identity-laden purchase semantics across all three tiers, the Cell B Japanese-cult tier is densely populated by named-tradition brands (Iwachu, Shun, Global, Tojiro) whose canonical-channel category authority varies independently of their cultural footprint, and the cross-language discourse environment (English category authority alongside Japanese-language craft canonization) exposes mention-detection conditions the program had not previously formalized. The two phases together carry the program’s earliest methodology lineage — pre-v1.4 single-channel Recall scoring under the original Phase B query battery — and together establish the constructs that subsequent phases refine.

v0.16 produced the original Regime 4 boundary case and the program’s first documented instance of substrate-language carryforward: Japanese-language category-bound discourse in the panel’s Recall responses returned different brand mention patterns than English-language responses for the same sub-

strate, with named-craft tradition brands appearing more readily in responses to category prompts framed in Japanese script than in English. The finding informed subsequent panel design and mention-detection rules: v1.4's canonical-detection specification adopted accent-stripped, word-boundary-anchored matching against multi-script-aware brand registries, in part to manage this carryforward effect at the scoring layer rather than at the prompt layer. The v0.16 H_IdentityLoad_moderator leg returned PARTIAL, with the three-cell IL-gradient signature visible in the v1.4 single-channel data but not strong enough to resolve to a CONFIRMED verdict under the protocol then in force. Regime 4 itself emerged as a methodology object in this phase — the substrate condition under which the IL-gradient moderator becomes evaluable — and the four-regime taxonomy that organized this condition was formalized in v1.2 (González Castro, 2026b) immediately after v0.16's acquisition.

v0.17 surfaced the program's first named dissociation case. Iwachu, a Japanese kitchen-knives brand from the v0.17 panel's Cell B, scored at the high end of the Recognition layer ($C_P \geq 5$ across the panel) while returning sparse canonical Recall — the pattern that would subsequently be formalized in v1.4 as the Iwachu dissociation type (high Recognition with low canonical Recall: $C_P \geq 5 \wedge R_{cat} \leq 2$). The brand's profile illustrates the dissociation mechanism: every panel intermediary correctly categorized Iwachu as a kitchen-knives brand at the Recognition layer (the LLM training distributions covered the brand's existence), but the canonical-channel Recall frames (best, recommended, highest-quality) returned the brand at substantially lower frequency than would be expected from its Recognition score, indicating that category authority in the panel's canonical-channel discourse did not track category-membership canonization in the way the v1.4 framework would later predict. v0.17's H_Regime4 returned FALSIFIED-on-panel-inadequacy: the Recognition layer for the v0.17 panel was insufficiently discriminating across the relevant within-cell variance condition, and the verdict resolved at the panel-adequacy boundary rather than at a substantive Regime 4 result. The v0.17 H_IdentityLoad_moderator leg propagated FALSIFIED-on-panel-inadequacy into the joint. The panel-inadequacy condition itself became a methodology object that v1.3 (González Castro, 2026c) developed into the Phase A pivot-validation specification — a procedure for diagnosing when a panel's Recognition discrimination falls below the threshold required for downstream Regime 4 evaluation, and for choosing between panel substitution and substrate-side adjustment in response.

Under v1.6 retrospective scoring, both kitchenware phases classify as differential by Increment 1 narrative — each carries within-cell C_P variance sufficient to reject the uniform-saturation rule. Increment 2 does not apply: R_{cult} is a v1.5 prospective increment first acquired at v0.20, and no defensible retrospective channel re-mapping is available for either kitchenware phase. Increment 3 is out of v1.6 retrospective scope. The kitchenware family contributes the Iwachu dissociation pattern (the empirical motivation for the v1.4 Recognition $\times$ Recall decomposition), the substrate-language-carryforward finding (a methodology object that constrains downstream panel design), and the original Regime 4 boundary definition. These three

contributions carry into §5's claim L2 (multi-component construct) and L4 (three dissociation quadrants).

4.2 Indie fragrance (v0.18)

The indie fragrance substrate (González Castro, 2026i) was the program's first English-language extension of the dissociation framework outside the kitchenware family. The substrate is structurally well-suited to IL-gradient testing: indie fragrance consumer-discovery operates through community-curated discourse environments — perfumistas forums, niche perfumery review channels, indie-fragrance newsletter ecosystems — where the canonical-channel category authority (anchored by professional perfume critics and editorial coverage) operates independently of the cultural footprint that drives cult-tier brand emergence. The panel was constructed across heritage / niche perfumery (Cell A — established perfume houses with established editorial recognition), perfumista-cult / DTC fragrance (Cell B — independent perfumers and small houses with strong community discourse), and mass-market fragrance (Cell C — designer-licensed and drugstore fragrance lines), with the IL-gradient design carrying forward from the kitchenware phases. v0.18 ran under v1.4 single-channel Recall scoring and produced the program's first multi-cell Iwachu generalization: the high-Recognition / low-canonical-Recall pattern surfaced in more than one cell, extending the construct from the Japanese-language anchor to an English-language perfumistas-native discourse environment and establishing that the dissociation pattern is not specific to a single substrate family or discourse language.

The Identity-Load moderator concept was first articulated as a coherent cross-phase hypothesis in v0.18, building on the IL-gradient signatures visible in v0.16 and v0.17. The v0.18 single-channel Recall data carried a clear cell-structured asymmetry — high-IL Cell B brands surfaced in the panel's Recall responses at different relative frequencies than medium-IL Cell A or low-IL Cell C brands — but the v1.4 framework's single-channel scoring could detect the asymmetry without decomposing it into its canonical and cultural components. The v0.18 asymmetry signal was the methodological motivation for the v1.5 two-channel Recall decomposition (González Castro, 2026e): if Recall behaves differently in canonical-authority frames than in cultural-footprint frames for high-IL substrate cells, then a protocol that scores Recall as a single mention count is conflating two empirically separable pathways. v1.5 was designed to test exactly this conflation in subsequent prospective phases. The v0.18 H_Regime4 returned PARTIAL, with the substrate's Phase A Recognition distribution producing a worst-cell modal share of 0.69 — differential, not uniform, but with one cell sufficiently close to saturation to constrain the within-cell discrimination available for downstream Recall analysis. The v0.18 H_IdentityLoad_moderator joint, evaluated as a three-leg combination over v0.16 × v0.17 × v0.18, returned PARTIAL, carrying the moderator construct into the subsequent prospective phases as a hypothesis worth measuring with the more discriminating Recall instruments that v1.5 would introduce. Figure 3 shows the Iwachu generalization pattern's cross-phase distribution, with

v0.18 marking its first multi-cell appearance.

Iwachu dissociation cases across the five-substrate anchor base

High Recognition ($C_P \geq 5$) with sparse canonical Recall ($R_{cat} \leq 2$) — multi-cell distribution in three of five families

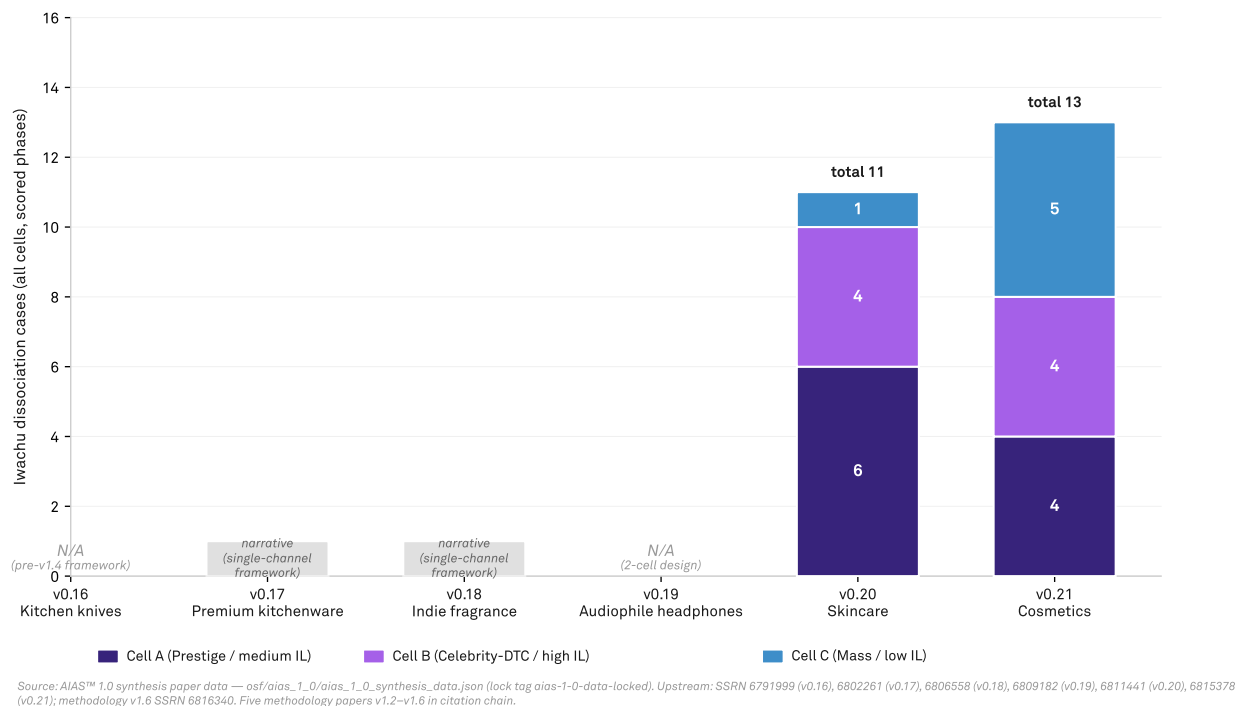


Figure 3: Iwachu dissociation cases across the five-substrate anchor base. High Recognition ($C_P \geq 5$) with sparse canonical Recall ($R_{cat} \leq 2$). Multi-cell distribution in three of five families.

Under v1.6 retrospective scoring, v0.18 classifies as differential by Increment 1 narrative — the modal share of 0.69 worst cell is well clear of the uniform-saturation threshold of 1.000. Increment 2 does not apply (R_{cult} unavailable). Increment 3 is out of retrospective scope. The indie fragrance phase contributes the substrate-family generalization argument (Iwachu beyond the Japanese-language anchor), the first IL-gradient signature articulated as a moderator framework, and the methodology motivation for v1.5's two-channel Recall decomposition. The Recall-asymmetry observation that v0.18 surfaced under v1.4 single-channel scoring is the methodology bridge from the v1.4 single-construct Recall framework to the v1.5 two-channel R_{cat} / R_{cult} decomposition that v0.20 and v0.21 would subsequently exploit; without v0.18's substrate-level evidence that the asymmetry was real and substrate-structured, the v1.5 two-channel design would have been speculative rather than empirically motivated.

4.3 Audiophile headphones (v0.19)

The audiophile headphones substrate (González Castro, 2026j) departs from the three-cell IL-gradient design in a structurally informative way. The substrate's consumer-discovery surface is densely populated by community-curated discourse — audiophile forums, headphone-review YouTube channels, community-canonization rituals that turn specific models and tunings into cult-tier reference points — that imposes high cultural footprint on every category-member brand from the entry tier to the heritage tier. A standard three-cell IL-gradient design fits the substrate poorly: the high-IL Cell B position that would normally separate cultural-cult brands from heritage Cell A brands is not available, because the heritage brands themselves carry cult-tier cultural footprint in the substrate's discourse environment. The v0.19 panel design accommodated this by using a two-cell Heritage / Boutique structure with uniform Identity Load by construction — a substrate condition the IL-gradient framework cannot test against, but one that is itself empirically informative about the limits of the framework's domain. Per the program's discipline of excluding ill-fitted phases from joint moderator verdicts rather than forcing them through, v0.19 was excluded from the H_IdentityLoad_moderator joint with explicit pre-registered rationale.

v0.19 ran under the v1.4.x protocol with the two-cell design and produced a Cell B saturation-prone Recognition distribution and a C2 FAILED → UNDETERMINED Regime 4 verdict under the v1.5 backport scoring. The Cell B modal share reached 0.875 — high but not the singleton 1.000 that would trigger uniform-saturation classification under the v1.6 Increment 1 rule, and high enough to constrain the within-cell variance the v1.5 C2 multi-statistic requires. The UNDETERMINED verdict is itself substantively informative: a substrate that does not satisfy the within-cell adequacy condition cannot be evaluated for Regime 4, but the unevaluable status is methodologically distinct from a substrate that fails the condition. The panel-inadequacy finding from v0.17 carried into v0.19 as a methodology constraint the substrate's design partially compensated for and partially exposed: the audiophile substrate's intrinsic Recognition saturation made within-cell variance scarce even with the Boutique tier deliberately constructed to introduce it, and the Phase A pivot-validation procedure that v1.3 had developed in response to v0.17 was load-bearing for managing the inadequacy condition at acquisition rather than discovering it post-hoc.

Under v1.6 retrospective scoring, v0.19 classifies as mixed by Increment 1 narrative — Cell B's high modal share approaches but does not satisfy the uniform-saturation rule (distinct C_P = 1 in all cells), and the Heritage cell carries enough variance to keep the substrate on the differential side of the classification boundary. Increment 2 does not apply: the v0.19 acquisition predates the v1.5 two-channel Recall design, and the single-channel q1-q6 frames used at acquisition cannot be retrospectively decomposed into R_cat and R_cult without introducing measurement artifacts not present at the acquisition layer. Increment 3 is out of retrospective scope. The audiophile phase contributes substrate-shape information — uniform-IL as a substrate configuration the IL-gradient framework cannot test against, and an empirical reference

point for the differential / mixed / uniform-saturation distinction that v1.6 Increment 1 formalizes. v0.19 is the program's nearest miss on the uniform-saturation rule: high modal share without singleton distinct count. v0.21's cosmetics case (§4.5) is the rule's first measured trigger, and the contrast between the two substrates' Phase A distributions is the empirical content of the rule's discrimination value. Without v0.19's near-miss reference point, the v0.21 uniform-saturation finding would be a singular observation rather than the resolution of a substrate-shape distinction the methodology had been tracking since v0.17.

4.4 Skincare (v0.20)

The skincare substrate (González Castro, 2026k) was the program's first prospective phase under v1.5's full sequence — multi-statistic C2, two-channel Recall, and the v1.4 dissociation framework operating jointly. The panel was constructed across heritage / clinical / dermatologist-anchored skincare (Cell A), celebrity-DTC skincare (Cell B), and mass / drugstore skincare (Cell C), with the IL-gradient design carrying forward from the indie fragrance phase. v0.20 produced the program's first two-channel acquisition and the first prospective application of the v1.5 C2 multi-statistic adequacy rule against a fresh substrate.

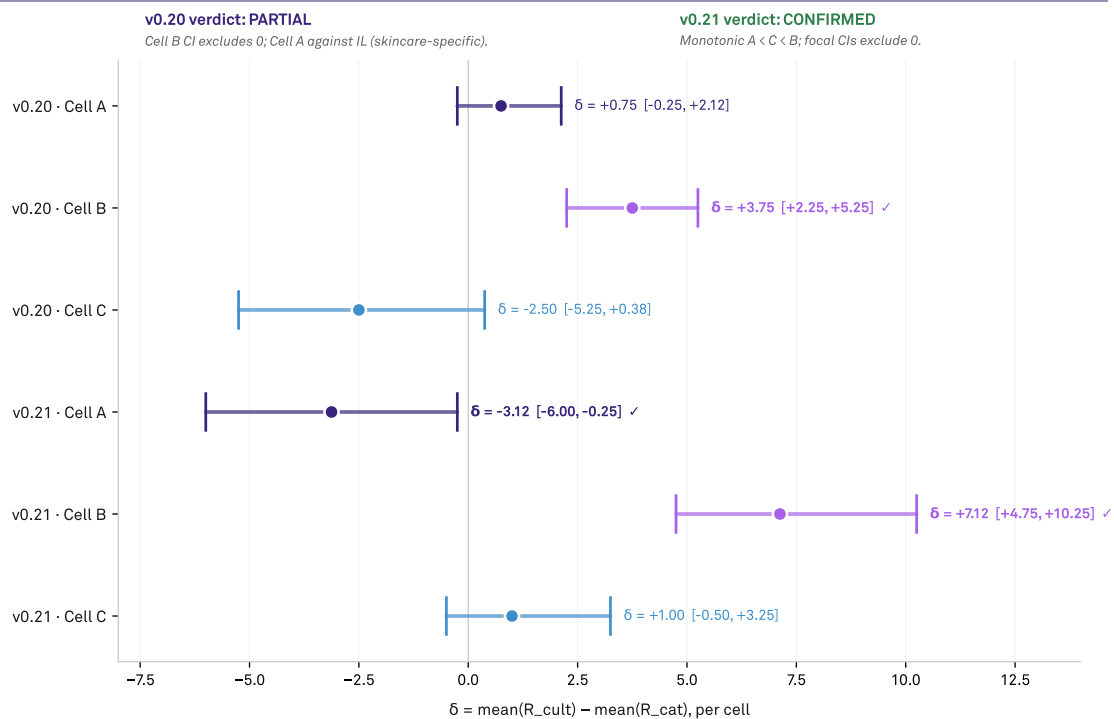
v0.20's headline findings are differential by design intent and substrate-specific in their structural details. H_Type2_emergence returned PARTIAL: Cell B Type 2 count was 2 (Rhode and Augustinus Bader), below the pre-registered EMERGED threshold of 3 but consistent with the IL-gradient prediction that high-IL cells should produce Type 2 cases. H_Regime4 returned FALSIFIED at C2: the Phase A Recognition distribution produced distinct C_P counts of 2 in Cell A, 3 in Cell B, and 1 in Cell C — three differently-shaped distributions in a single substrate, with the Cell C saturation driving the within-cell C2 failure and the Cell A bimodality producing the within-cell variance signal v1.5 was designed to surface. H_IdentityLoad_moderator returned NARROWED on the five-leg joint, with the within-phase IL signature carrying directionally with prediction but the Cell A canonical pole shifting from prestige (the cosmetics analogue) to mass (the skincare-specific finding): clinical / heritage skincare brands such as CeraVe, La Roche-Posay, and Eucerin do not concentrate in the canonical-channel Recall the way prestige cosmetics brands do, because the skincare substrate's canonical authority lives in dermatologist-recommended and clinical-result-anchored discourse rather than in editorial-prestige curation, and the substrate's IL-gradient signal accordingly lives in the Cell B versus Cell C contrast rather than the Cell A versus Cell B contrast that the standard prediction frames. The finding is substrate-specific rather than methodology-disqualifying: the IL-gradient model holds for the substrate, but the cell with the strongest canonical-channel lead is not the one the standard panel-design heuristic predicts.

Under v1.6 retrospective scoring (González Castro, 2026f), v0.20 classifies as differential by Increment 1 (distinct C_P = 2, 3, 1 across cells; the uniform-saturation rule rejects the substrate unambiguously). Increment 2 returns PARTIAL on the H_IdentityLoad_Direct pathway: Cell B $\delta = +3.75$ with 95% bootstrap CI

[+2.25, +5.25] excludes zero in the IL-predicted direction; Cell A $\delta = +0.75$ with CI $[-0.25, +2.12]$ is against the IL-predicted direction with CI overlapping zero; Cell C $\delta = -2.50$ with CI $[-5.25, +0.38]$ carries the substrate's R_cat dominance in the mass tier and breaks the monotonic-gradient check with Cell A. Increment 3 is out of v1.6 retrospective scope: the cross-validated-emergent source for the Phantom reference vocabulary is the substrate itself, a circularity the v1.6 construction rule was designed to avoid. Figure 5 shows the IL Direct δ values with 95% bootstrap CIs across both v0.20 and v0.21 cells.

H_IdentityLoad_Direct — per-cell $\delta = \text{mean}(R_{\text{cult}}) - \text{mean}(R_{\text{cat}})$ with 95% bootstrap CIs

v0.20 PARTIAL (skincare; Cell B vs. Cell C signal) → v0.21 CONFIRMED (cosmetics; monotonic A < C < B)



Source: AIAS™ 1.0 synthesis paper data — osf/aias_1.0/aias_1.0_synthesis_data.json (lock tag aias-1-0-data-locked). Upstream: SSRN 6791999 (v0.16), 6802261 (v0.17), 6806558 (v0.18), 6809182 (v0.19), 6811441 (v0.20), 6815378 (v0.21); methodology v1.6 SSRN 6816340. Five methodology papers v1.2–v1.6 in citation chain.

Figure 4: H_IdentityLoad_Direct per-cell $\delta = \text{mean}(R_{\text{cult}}) - \text{mean}(R_{\text{cat}})$ with 95% bootstrap CIs. v0.20 PARTIAL (skincare; Cell B vs. Cell C signal) → v0.21 CONFIRMED (cosmetics; monotonic A < C < B).

A cross-phase synthesis-layer contribution surfaces here. Applied uniformly to the v0.20 acquisition data, the all-cell algorithmic dissociation framework — the same $R_{\text{cat}} \leq 2 \wedge R_{\text{cult}} \geq 5$ rule that v0.20's H_Type2_emergence verdict matrix correctly anchored to Cell B within v0.20's own scope — recovers a third Type 2 case the verdict matrix did not surface: La Mer (Cell A — Prestige; C_P = 6, R_cat = 2, R_cult = 7). La Mer's Cell A position places her outside the v0.20 verdict's Cell-B-anchored evidence base, which is why the published PARTIAL verdict (correctly) did not include her. Read across the cross-phase view, La Mer is the first algorithmic out-of-cell Type 2 case in the program. The v0.20 published verdict stands as the

authoritative within-phase result; the synthesis adds the cross-phase algorithmic perspective alongside it. v0.20 contributes the first two-channel decomposition acquisition, the skincare-specific Cell A architecture finding that informs the v1.6 Increment 2 verdict matrix's substrate-aware design, and (under cross-phase view) the first out-of-cell Type 2 anchor.

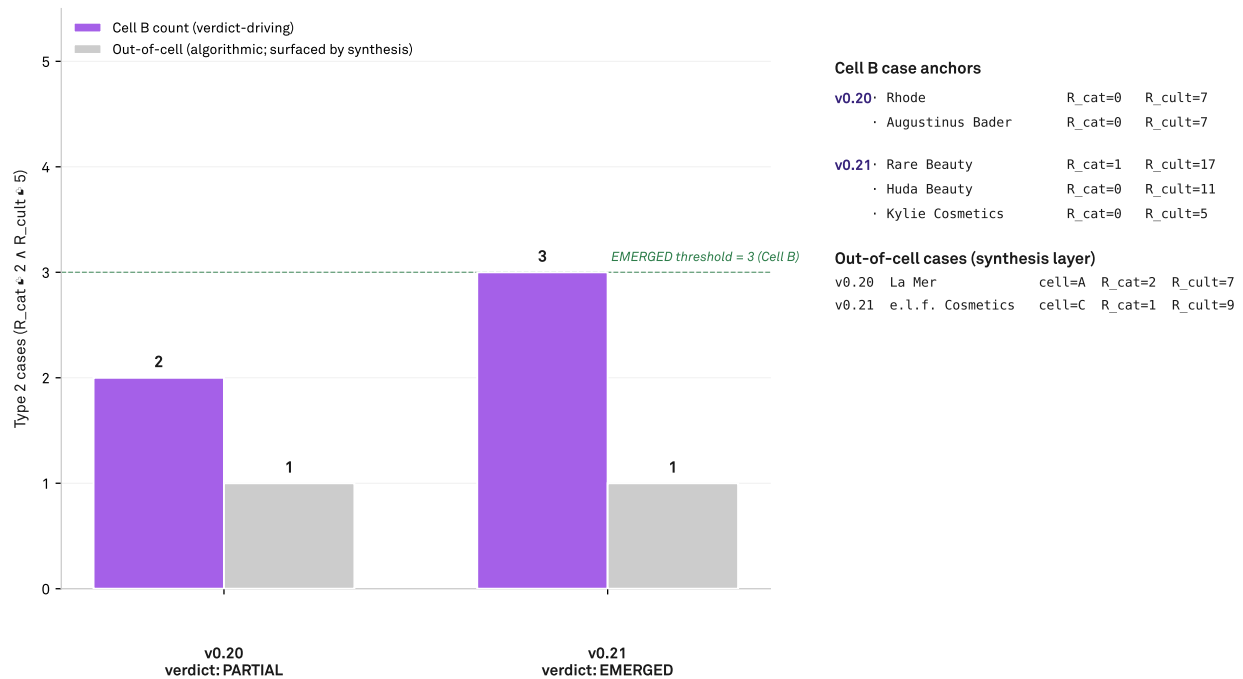
4.5 Cosmetics (v0.21)

The cosmetics substrate (González Castro, 2026l) is the program's strongest empirical anchor and the synthesis paper's headline finding-base. The panel was constructed across prestige (Cell A — medium IL: MAC Cosmetics, NARS, Bobbi Brown, Tom Ford Beauty, Giorgio Armani Beauty, Hourglass, Chantecaille, Laura Mercier), celebrity-DTC / cult (Cell B — high IL: Rare Beauty, Fenty Beauty, Haus Labs, Pat McGrath Labs, Charlotte Tilbury, Huda Beauty, Kylie Cosmetics, Anastasia Beverly Hills), and drugstore / mass (Cell C — low IL: Maybelline, L'Oréal Paris, CoverGirl, Revlon, NYX Professional Makeup, e.l.f. Cosmetics, Wet n Wild, Milani Cosmetics). v0.21 was the first phase prospectively pre-registered to test Type 2 emergence in a high-IL substrate cell against an EMERGED threshold of 3 Cell B cases.

v0.21's headline findings cleared the Type 2 EMERGED threshold and produced the program's strongest IL-gradient channel-asymmetry signal to date. H_Type2_emergence returned EMERGED with three Cell B cases: Rare Beauty (R_cat = 1, R_cult = 17 — the program's textbook Recognition × Recall dissociation case with C_P = 6 in the Recognition layer), Huda Beauty (R_cat = 0, R_cult = 11), and Kylie Cosmetics (R_cat = 0, R_cult = 5). H_Regime4 returned FALSIFIED at C2 under the v1.5 routing, with the Phase A distribution producing uniform Recognition saturation across all three cells (every panel brand scored C_P = 6/6; distinct C_P = 1 and modal share = 1.000 in Cell A, Cell B, and Cell C). H_Dissociation_substrate_generalization returned GENERALIZED with 13 lwachu cases distributed three-of-three across the panel cells — the program's largest single-phase lwachu count. Figure 4 shows the Type 2 cross-phase transition from v0.20 PARTIAL to v0.21 EMERGED with both the Cell-B-verdict-driving and all-cell algorithmic counts.

Type 2 quadrant emergence — v0.20 PARTIAL → v0.21 EMERGED

Cell B count drives the H_Type2_emergence verdict (≥ 3 → EMERGED); cross-phase all-cell count surfaces out-of-cell cases



Source: AIAS™ 1.0 synthesis paper data — osf/aias.1.0/aias.1.0_synthesis_data.json (lock tag aias-1.0-data-locked). Upstream: SSRN 6791999 (v0.16), 6802261 (v0.17), 6806558 (v0.18), 6809182 (v0.19), 6811441 (v0.20), 6815378 (v0.21); methodology v1.6 SSRN 6816340. Five methodology papers v1.2–v1.6 in citation chain.

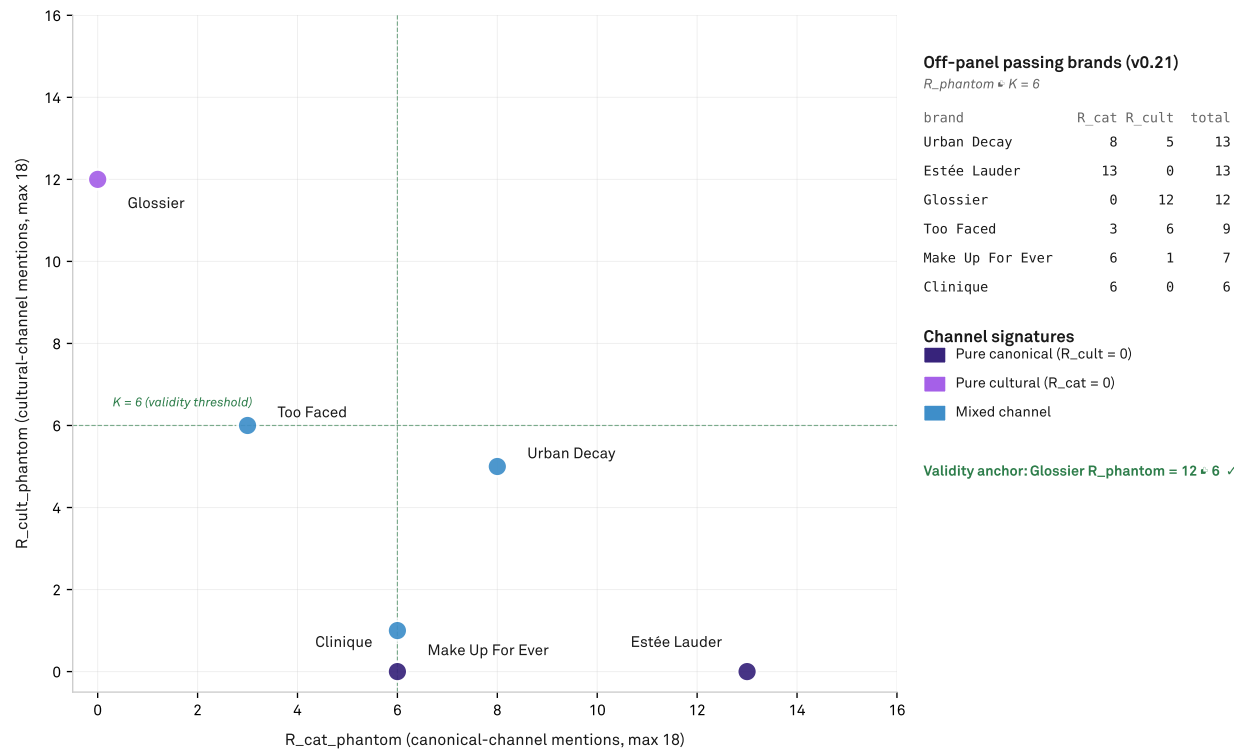
Figure 5: Type 2 quadrant emergence: v0.20 PARTIAL → v0.21 EMERGED. Cell B count drives the H_Type2_emergence verdict (≥ 3 → EMERGED); cross-phase all-cell count surfaces out-of-cell cases.

Under v1.6 retrospective scoring (González Castro, 2026f), v0.21 classifies as uniform-saturation by Increment 1 — the program’s first measured trigger of the rule, routing Regime 4 to REGIME-4-UNAVAILABLE-AT-RECOGNITION distinct from FALSIFIED. Increment 2 returns CONFIRMED on the H_IdentityLoad_Direct pathway: Cell B $\delta = +7.13$ with 95% bootstrap CI $[+4.75, +10.25]$ excludes zero in the IL-predicted direction (the program’s strongest cult-channel-lead signal); Cell A $\delta = -3.13$ with CI $[-6.00, -0.25]$ excludes zero in the opposite direction as the IL-gradient model predicts for the medium-IL prestige cell; Cell C $\delta = +1.00$ with CI $[-0.50, +3.25]$ satisfies the monotonic-gradient check (Cell A $\delta < \text{Cell C } \delta < \text{Cell B } \delta$). v0.21 is the program’s first CONFIRMED moderator verdict at any layer. Increment 3 returns CONFIRMED on H_PhantomBrandPersistence: six off-panel reference-vocabulary brands cleared $K = 6$ (Urban Decay 13, Estée Lauder 13, Glossier 12, Too Faced 9, Make Up For Ever 7, Clinique 6), and the validity anchor passed with margin (Glossier $R_{\text{phantom}} = 12 \geq \text{required } 6$). The phantom cohort exhibits a striking channel signature: three of the six are channel-pure, with Estée Lauder and Clinique appearing exclusively in canonical-channel frames ($R_{\text{cult}} = 0$ for both) and Glossier appearing exclusively in cultural-channel frames ($R_{\text{cat}} = 0$). The channel signatures track the brands’ Identity Load — prestige / heritage cosmetics in the canonical chan-

nel, celebrity / DTC / cult in the cultural channel — establishing that the IL-gradient mechanism operates substrate-wide rather than panel-internally. Figure 6 shows the phantom channel signature scatter.

v0.21 Phantom Brand Persistence — off-panel channel signature ($R_{cat_phantom} \times R_{cult_phantom}$)

Six off-panel brands clear $K = 6$; channel signatures track Identity Load (validity anchor Glossier passes with margin)



Source: AIAS™ 1.0 synthesis paper data — osf/aiaas_1_0/aiaas_1_0_synthesis_data.json (lock tag aiaas-1-0-data-locked). Upstream: SSRN 6791999 (v0.16), 6802261 (v0.17), 6806558 (v0.18), 6809182 (v0.19), 6811441 (v0.20), 6815378 (v0.21); methodology v1.6 SSRN 6816340. Five methodology papers v1.2–v1.6 in citation chain.

Figure 6: v0.21 Phantom Brand Persistence — off-panel channel signature ($R_{cat_phantom} \times R_{cult_phantom}$). Six off-panel brands clear $K = 6$; channel signatures track Identity Load. Glossier validity anchor passes with margin.

The cross-phase synthesis-layer contribution flagged in §4.4 (La Mer in v0.20 as the first algorithmic out-of-cell Type 2 case) continues here. The all-cell algorithmic dissociation framework, applied uniformly to v0.21 data, recovers a fourth Type 2 case the v0.21 verdict matrix correctly did not surface within its own Cell-B-anchored scope: e.l.f. Cosmetics (Cell C — Drugstore / mass; $C_P = 6$, $R_{cat} = 1$, $R_{cult} = 9$). e.l.f. is the second algorithmic out-of-cell Type 2 case in the program, surfacing in the first phase prospectively pre-registered to anticipate Type 2 emergence in a high-IL substrate cell. Together with La Mer in v0.20, e.l.f. jointly establishes out-of-cell Type 2 as a recurrent feature of the dissociation framework that per-phase Cell-B-anchored verdict matrices cannot surface within their own scope — different substrate, different cell, same dissociation pattern. The v0.21 published verdict stands as the authoritative within-phase result; the synthesis adds the cross-phase algorithmic perspective alongside it. v0.21 contributes the Type 2 EMERGED

anchor, the program's first CONFIRMED IL-gradient moderator verdict, the operationalization of Phantom Brand Persistence as a measured component, the program's first measured uniform-saturation Recognition state, and (under cross-phase view) the second out-of-cell Type 2 anchor that closes the synthesis-layer pattern with v0.20 La Mer.

5. The AIAS™ Presence construct

The construct claim is developed in seven layers below. Each subsection states the claim layer, defines what support and boundary mean for that layer, and names the empirical anchors in §4 that carry the layer.

5.1 L1 — AI Availability as a third measurable layer

AI Availability — the brand-level probability that an AI intermediary retrieves, recommends, or selects a brand in a category-anchored decision context — is a third measurable layer of brand availability alongside Ehrenberg-Bass Mental Availability and Physical Availability. Measurability here is the operative claim: under the AIAS™ Presence Measurement Protocol's pre-registered measurement procedure, AI Availability resolves to falsifiable verdicts (CONFIRMED, PARTIAL, FALSIFIED, EMERGED, GENERALIZED, NARROWED, REGIME-4-UNAVAILABLE-AT-RECOGNITION) that can be evaluated against each phase's locked verdict matrix and replicated across substrate families. The empirical support for L1 is the full six-phase corpus: all five substrate families (§4.1 through §4.5) produced coherent verdicts under the protocol, and no substrate has driven the program to abandon the construct or rebuild the measurement procedure at the foundational layer. The boundary is the demarcation from construct validity — measurability is established here; predictive, convergent, and discriminant validity remain Phase 3 work.

5.2 L2 — Multi-component construct (Recognition × Recall)

AI Availability decomposes into at least two empirically separable components: Recognition (C_P, the brand's category-membership canonicity across the panel) and Recall (the brand's retrieval into category-anchored open-ended responses). The decomposition is empirically motivated, not theoretical: v0.17's first Iwachu case (§4.1) demonstrated that a brand can be fully recognized at the Recognition layer (C_P ≥ 5) while returning sparse canonical Recall, dissociating the two components empirically; v0.18's multi-cell Iwachu generalization (§4.2) extended the dissociation across cells; v0.21's uniform Recognition saturation (§4.5) confirmed the dissociation in the limit case where Recognition is exhausted at the substrate level and Recall remains the construct's discrimination signal. The v1.4 methodology specification (González Castro, 2026d) formalizes the decomposition and is the protocol's load-bearing methodology object at this layer.

5.3 L3 — Two-channel Recall ($R_{cat} \times R_{cult}$)

Recall decomposes further into a canonical-authority channel (R_{cat} , anchored by best, recommended, and highest-quality frames) and a cultural-footprint channel (R_{cult} , anchored by popular, celebrity, and viral / cult frames). The two channels operate on disjoint inputs and can be managed independently. Empirical support spans the v1.5 prospective phases. v0.18's substrate-level Recall asymmetry under v1.4 single-channel scoring (§4.2) was the methodological motivation; v0.20's first two-channel acquisition (§4.4) supplied the first prospective evidence; v0.21's Rare Beauty case (§4.5) — $R_{cat} = 1$, $R_{cult} = 17$ against a Recognition score of $C_P = 6$ — anchors the textbook two-channel dissociation case in the program. The v1.5 methodology specification (González Castro, 2026e) formalizes the channel decomposition.

5.4 L4 — Three dissociation quadrants

Three canonical patterns occupy the Recognition $\times$ Recall plane: Iwachu ($C_P \geq 5 \wedge R_{cat} \leq 2$), Type 1 ($R_{cat} \geq 5 \wedge R_{cult} \leq 2$), and Type 2 ($R_{cat} \leq 2 \wedge R_{cult} \geq 5$). Iwachu generalizes across all five substrate families with multi-cell distribution in three of them (kitchenware §4.1, indie fragrance §4.2, cosmetics §4.5; and at the scored level in v0.20 §4.4 and v0.21 §4.5). Type 1 surfaces in the v0.20 and v0.21 prestige cells (§4.4 and §4.5). Type 2 cleared the EMERGED threshold in v0.21 after PARTIAL in v0.20, with Cell B counts of 3 and 2 respectively against the pre-registered threshold of 3 for EMERGED. The cross-phase synthesis-layer contribution flagged in §4.4 and §4.5 — La Mer in v0.20 (Cell A) and e.l.f. Cosmetics in v0.21 (Cell C) jointly establishing out-of-cell Type 2 as a recurrent feature of the dissociation framework — carries L4 beyond the per-phase Cell-B-anchored verdict view into a substrate-wide dissociation pattern that the per-phase view cannot surface within its own scope.

5.5 L5 — Identity-Load moderator (channel asymmetry tracks IL)

Per-cell channel asymmetry $\delta = \text{mean}(R_{cult}) - \text{mean}(R_{cat})$ tracks the Identity-Load gradient of the substrate's cell construction. High-IL cells produce $\delta > 0$ (cultural-channel-leading); medium-IL cells produce $\delta < 0$ (canonical-channel-leading); low-IL cells produce δ values bounded by the monotonic-gradient check between the focal cells. Empirical support: v0.21 $H_IdentityLoad_Direct$ CONFIRMED (§4.5) is the program's first CONFIRMED moderator verdict at any layer, with Cell A $\delta = -3.13$, Cell B $\delta = +7.13$, and Cell C $\delta = +1.00$ satisfying the monotonic ordering $A < C < B$; v0.20 $H_IdentityLoad_Direct$ PARTIAL (§4.4) carries the moderator directionally in Cell B while surfacing the skincare-specific Cell A architecture that the substrate-aware verdict matrix accommodates; v0.18's articulation (§4.2) anchors the hypothesis at the cross-phase joint layer. The v1.6 Increment 2 specification (González Castro, 2026f) formalizes the test. The boundary of L5 is important: Identity Load is operationalized through cell construction at the panel-design layer, not measured at the consumer layer; the moderator's mechanism is the IL gradient of the substrate's panel design

as it interacts with the two-channel Recall structure of the panel intermediaries.

5.6 L6 — Phantom Brand Persistence as an empirical regularity

Off-panel brands persistently surface in panel-anchored category retrieval. The phenomenon is framed in the Ehrenberg-Bass tradition as an empirical regularity — a stable pattern recoverable through replication-discipline measurement, comparable in framing to Double Jeopardy or the duplication-of-purchase law — rather than as a coordinate mechanism deduced from a first-principles model. Empirical support: v0.21 H_PhantomBrandPersistence CONFIRMED (§4.5) with six off-panel brands clearing $K = 6$, the Glossier validity anchor passing with margin ($R_{\text{phantom}} = 12 \geq 6$), and a channel signature in which three of the six phantom brands are channel-pure (Estée Lauder and Clinique exclusively canonical; Glossier exclusively cultural) tracking the brands' Identity Load. The v1.6 Increment 3 specification (González Castro, 2026f) formalizes the construct with an exogenous reference-vocabulary construction that breaks the endogeneity hazard of discovery-driven vocabulary expansion. The boundary at L6 is calibration vs. generalization: v0.21 is the calibration anchor for the construct (the substrate where $K = 6$ was anchored and where the validity check was first applied), and v0.22+ prospective phases under v1.6 are the genuine generalization test. The L6 claim is operationalization-and-confirmed-in-calibration, not yet replicated-across-substrates.

5.7 L7 — AIAS™ 1.0 as Presence, with the six-component composite as a multi-year roadmap

The synthesis paper claims measurement of the Presence component of the AIAS construct, not the full multi-component composite. The Tri-System framework (González Castro, 2026a) names the full composite as a 0–100 score across six measurable dimensions — Presence, Ranking, Consistency, Coverage, Grounding, Sentiment — and positions the composite as a multi-year research arc. AIAS™ 1.0 operationalizes Presence; the remaining five dimensions ship under a Phase 4 successor program. The naming convention is load-bearing: AIAS™ 1.0 does not claim measurement of the full composite, and the present paper's construct claim does not extend past the Presence layer. The honest framing is the program's credibility asset. Measurability of one component, established under pre-registration discipline against a five-substrate anchor base, is a stronger claim to make and defend than measurement of an under-specified composite. Reviewers can audit the present surface area against the present evidence; the multi-year roadmap is signposted as future work, not as a deferred liability against the present claim. The roadmap's specification is reserved for Phase 4 pre-registration discipline and is not load-bearing for L1 through L6 above. The version number itself encodes the discipline: 1.0 names the moment when the construct's first component reaches a falsifiable measurement procedure under cross-substrate replication, and the version-number arc that follows will increment as the construct's additional components reach the same standard. The construct does not jump to 2.0 by claiming additional components without measuring them; the version increments track

the measurement surface, not the architectural ambition.

6. Discussion

6.1 Implications for marketing science

The construct claim is additive to the Ehrenberg-Bass canon, not a replacement of Mental or Physical Availability. Sharp's two-layer formulation (Sharp, 2010; Romaniuk, 2018; Sharp & Romaniuk, 2021) remains the discipline's central availability framework; the present synthesis adds a third measurable layer at the AI-mediated retrieval surface that the two-layer formulation does not cover by construction. The additive framing matters for downstream integration. Categories whose purchase environments are mediated by AI intermediaries — increasingly the default consumer-discovery surface across consumer categories — gain a third construct that brand managers and marketing scientists can include alongside Mental Availability metrics (category entry points, distinctive asset strength, prompted brand recall) and Physical Availability metrics (distribution depth, channel coverage, on-shelf presence). The discipline's measurement tradition has long accommodated additive constructs that operate on disjoint inputs from existing constructs; AI Availability is the natural addition for the AI-mediated portion of the purchase environment. The synthesis enters the discipline under the same pre-registration discipline that the Ehrenberg-Bass tradition has built credibility on — locked panels, locked hypothesis matrices, public data and code deposits, replication across substrates — and inherits the credibility infrastructure of the tradition rather than starting from outside it. Integration of the third layer into existing brand-tracking instruments is a Phase 3 task: cross-construct calibration (where do Mental Availability and AI Availability covary, where do they dissociate?), shared-substrate measurement designs, and convergent / discriminant validity testing against the existing two constructs will determine the integration's operational form. The present synthesis establishes the construct as measurable; the discipline's integration of the construct into the canonical two-layer framework is the work the next decade of measurement-program research will do.

6.2 Implications for brand practice

For brand practice, the construct's most actionable finding is L3's two-channel Recall decomposition. R_{cat} and R_{cult} operate on disjoint inputs and can be managed separately. Canonical-channel Recall responds to authority surfaces — editorial coverage, expert recommendation, category-best curation, professional certification, dermatologist endorsement, makeup-artist recommendation, sommelier rating. Cultural-channel Recall responds to discourse density — social-media volume, celebrity endorsement, viral content, community-curated cult-tier discourse, founder-identity association. The v0.21 cosmetics evidence (§4.5) establishes that brands can occupy substantially different positions on the two channels while remaining within a single category, and that the IL-gradient of the substrate's panel design predicts

where in the two-channel space a brand will land. Practitioners can audit a brand's current channel position with the protocol's measurement procedure and target the underperforming channel with the appropriate intervention class without conflating the two pathways into a single Recall budget. The Rare Beauty case at §4.5 ($R_{cat} = 1$, $R_{cult} = 17$ against $C_P = 6$) is the textbook diagnostic: a brand with full Recognition and overwhelming cultural-channel Recall but negligible canonical-channel Recall has identifiably different brand-strategy options than a brand with the opposite profile, and the synthesis's measurement procedure surfaces the distinction at the audit layer rather than leaving it to interpretation. L6's Phantom Brand Persistence layer carries a second practical finding: brand presence in AI-mediated retrieval can exceed the boundaries of any single panel construction. A brand that does not appear in a measurement panel can nonetheless persistently surface in the panel's retrieval responses, with channel signature tracking the brand's Identity Load. The phantom layer is treated as an empirical regularity rather than a coordinate mechanism — the synthesis claims that the pattern reliably recurs, not that any particular brand's phantom presence is causally explained by any particular brand-strategy intervention. For brands at the substrate's edge — strong cultural footprint but not on the canonical panel of any single category-tracking study — the phantom layer is the reason their AI Availability cannot be inferred from panel-internal absence; it must be measured against an exogenously-constructed reference vocabulary at the substrate level.

6.3 Limitations

Several limitations bound the present synthesis. The reference panel is held fixed at the v0.17 six-slot specification across phases, and provider model substitutions over future-phase horizons are expected; v0.21's verdicts are properties of brand $\times$ substrate $\times$ panel triples measured at acquisition time, and substrate verdicts may shift as the underlying intermediaries are updated by their providers. The fixed-panel discipline is the program's cross-phase comparability anchor, but it is also a measurement-window constraint that the construct's stability across model generations remains future-work to demonstrate. Four of the five substrate families operated in English-language consumer-discovery environments (the kitchenware family's substrate-language carryforward at §4.1 is the only cross-language acquisition in the program), and cross-language replication is held as future work — the present synthesis claims the construct is measurable in the substrates measured, not that the measurement procedure transfers unchanged across discourse-language boundaries. Cell-classification limits surfaced through the e.l.f. Cosmetics out-of-cell Type 2 case in §4.5: the IL-gradient cell structure assumes brands behave per their supply-side tier classification, and a brand that crosses the supply-side / discourse-side boundary on observable channels exposes friction in the classification scheme. The retrospective-scoring boundary itself is a methodology constraint: v0.16 through v0.19 do not carry R_{cult} data and therefore cannot be evaluated against the v1.6 Increment 2 specification under the program's no-retrospective-channel-re-mapping discipline; the synthesis's $H_{IdentityLoad_Direct}$ evidence base is bounded to v0.20 and v0.21. The synthesis does not claim construct validity — predic-

tive validity against behavioral outcomes, convergent validity across alternative panel constructions, and discriminant validity against Mental and Physical Availability are Phase 3 work and are not established here. The synthesis is silent on the consumer-behavior correlate of AI Availability; the correlation is plausible on theoretical grounds but is not measured by the present program, and any reading of the synthesis that imports a behavioral-correlate claim is reading beyond the evidence the paper presents.

7. Future research

7.1 Phase 3 — construct validation

The most consequential next step is construct validation. Predictive validity against behavioral outcomes — whether AI Availability scores predict downstream consumer behavior such as brand consideration, search behavior, or purchase — is the discipline’s primary validation criterion and the program’s Phase 3 anchor. The Phase 3 design requires a behavioral-outcome dataset paired with AI Availability scores measured at the same brand-substrate boundary, against panel constructions held fixed for measurement comparability and varied for convergent-validity testing. Discriminant validity against Mental Availability and Physical Availability — establishing that AI Availability captures variance the existing constructs do not — requires paired measurements of all three constructs against a common brand-substrate cohort. Both predictive and discriminant validity sit in the Phase 3 pre-registration roadmap and are not in the present synthesis’s claim scope. The validation design will inherit the same pre-registration discipline that v1.2 through v1.6 established for the measurement layer: hypothesis matrices locked ex-ante at git tag, panel and substrate registries deposited at OSF, scoring code released alongside results, and verdict matrices specified independently of the data the validation will measure. Construct validity is the standard the AI Availability construct must clear before the discipline integrates it alongside the canonical two layers; the present synthesis’s role is to establish the measurability claim that makes the validation work tractable.

7.2 Phase 4 — full six-component AIAS composite

The Tri-System framework (González Castro, 2026a) names the full AIAS composite as a six-component construct spanning Presence (operationalized in the present paper), Ranking, Consistency, Coverage, Grounding, and Sentiment. The remaining five components ship under a Phase 4 successor program at the intent level only; their specifications, verdict matrices, and pre-registration discipline are reserved for the Phase 4 pre-registration round and are not load-bearing for the present construct claim. The Phase 4 arc is multi-year; component-level prioritization, panel-design implications, and methodology successors will follow the same discipline that v1.2 through v1.6 have established for the Presence layer. The version-number arc above (AIAS™ 1.0 here) increments only when each successor component reaches falsifiable measurement against a cross-substrate anchor base — the discipline binds the version numbering to the measurement

surface rather than to architectural ambition, and the composite reaches a fully measured 6.0 state only when all six components have been operationalized to the standard the Presence component reaches in the present synthesis.

7.3 Substrate and language expansion

The five-substrate anchor base is the foundational evidence base for the construct claim, but the construct is not substrate-exhausted by the five families. v0.22 and subsequent prospective phases under v1.6 will extend the anchor base into additional substrate families (selection criteria to be locked at each phase's pre-registration), test the v1.6 increments prospectively against substrates not in the present retrospective scope, and refine the substrate-classification taxonomy as new substrate shapes surface. The substrate-classification taxonomy that v1.6 Increment 1 introduces (differential / mixed / uniform-saturation) is itself a methodology arc that the v0.22+ phases will populate: each new substrate either fits cleanly into one of the existing classes or surfaces a configuration the taxonomy has not yet encountered, and either outcome is empirically informative about the methodology's domain. Cross-language replication is the second expansion axis: four of the five present substrates operated in English-language consumer-discovery environments, and non-English-language substrate phases (Japanese, Spanish, Mandarin, and others to be selected on substrate-availability grounds) will test whether the construct's measurement procedure transfers across discourse-language boundaries with the panel-internal scoring discipline held fixed. Panel-construction sensitivity is the third expansion axis: studies that hold substrate and methodology version fixed while varying the panel — substituting providers, varying panel size, testing single-provider panels against multi-provider panels — will supply the convergent-validity evidence base that Phase 3 construct validation requires. The three expansion axes together carry the construct's empirical anchor base from the present five families into a sustained replication program under the discipline that the present synthesis establishes. The anchor base does not need to be exhaustive to be load-bearing; it needs to be deep enough that the construct's claims are testable, replicable, and falsifiable. The synthesis paper's role is to establish that depth has been reached; the expansion program's role is to extend it.

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Declarations

Declaration of interest. The author is employed by Samsung Electronics America (Director, Corporate Brand Creative and Governance). The AIAS™ Presence Measurement Protocol and the synthesis work reported here are the author's independent academic research, conducted outside the scope of employment, in the author's role as faculty at the School of Visual Arts MPS Branding Program and founder of Third System. Samsung had no role in the design, execution, analysis, or interpretation of this work. Each phase in the v0.16–v0.21 corpus included a pre-acquisition COI screen confirming no brand in the phase panel is affiliated with Samsung Electronics America; the synthesis introduces no new panel and inherits the per-phase COI screens as documented at each phase's osf.io/ec6wh/v<NN>/ deposit.

Funder. Self-funded.

Ethics. Not applicable; no human subjects. The research uses publicly accessible LLM APIs queried with non-personal, category-anchored prompts.

Data and code availability. All pre-registration artifacts, acquisition data, scoring code, retrospective scoring outputs, cross-phase synthesis data, chart sources, and verdict outputs are deposited under Open Science Framework project ec6wh at osf.io/ec6wh/aias_1_0/. The synthesis is locked at git tags `aias-1-0-outline-locked` (outline at r1), `aias-1-0-data-locked` (cross-phase source-of-truth data and JSON), `aias-1-0-charts-locked` (figures), and `aias-1-0-paper-locked` (paper at submission). The phase-level deposits at osf.io/ec6wh/v16/ through osf.io/ec6wh/v21/ carry each phase's pre-registration, acquisition data, scoring code, and verdicts as the upstream record; the methodology paper deposits at osf.io/ec6wh/methodology/v1_2/ through osf.io/ec6wh/methodology/v1_6/ carry the locked methodology versions.

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